

Table 1: Popular tools for Web application testing

Tool	URL
Vega	https://subgraph.com/vega/
Canoo Web Test	http://webtest.canoo.com/
Webrat	https://github.com/brynary/webrat
Watir	http://watir.com/
SoapUI	https://www.soapui.org/
Cucumber	https://cucumber.io/z
Selendroid	http://selendroid.io/
Selenium	https://www.seleniumhq.org/
Pylot	https://code.google.com/archive/p/pylt/downloads
OpenSTA	http://opensta.org/
Grinder	http://grinder.sourceforge.net/
Gatling	https://gatling.io/
Multi-Mechanize	https://multi-mechanize.readthedocs.io/en/latest/
Tsung	http://tsung.erlang-projects.org/
WebLoad	https://www.radvist.com/webload-download/
TestNG	https://testng.org
Locust	https://locust.io/
Capybara	https://github.com/teamcapybara/capybara
Apache JMeter	https://jmeter.apache.org/
Appium	http://appium.io/
FitNesse	http://fitnesse.org/

Nowadays, most of the smartphone apps are available on Android operating systems. They are widely adopted by mobile phone users because of their popularity and availability as open source distributions. Installation of Android apps in the Android Application Package (APK) format requires specific permissions, which are generally ignored by Android users.

With the increasing number of smartphones and mobile applications, there is a need to check the vulnerabilities of these devices to avoid misuse and exploitation.

There are a number of cloud repositories, from where the Android APK files are downloaded:

<https://apkpure.com/>
<https://www.apkmirror.com/>
<https://www.androiddrawer.com/>
<https://apk-dl.com/>
<https://en.aptoide.com/>
<https://androidapksfree.com/>
<https://www.appsapk.com/>
<https://apk4all.com/>

...and many others.

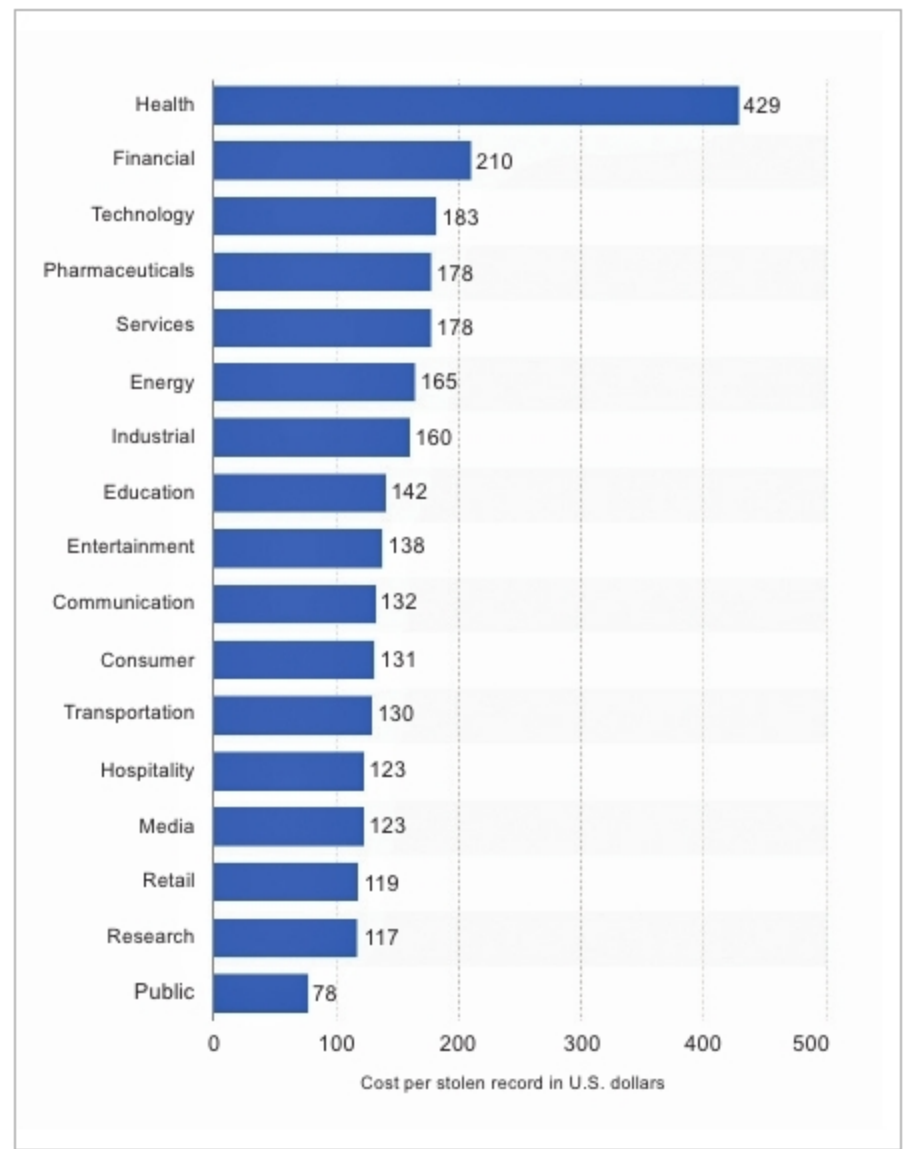


Figure 1: The cost implications of stolen data (US\$)
(Statista: <https://www.statista.com/>)

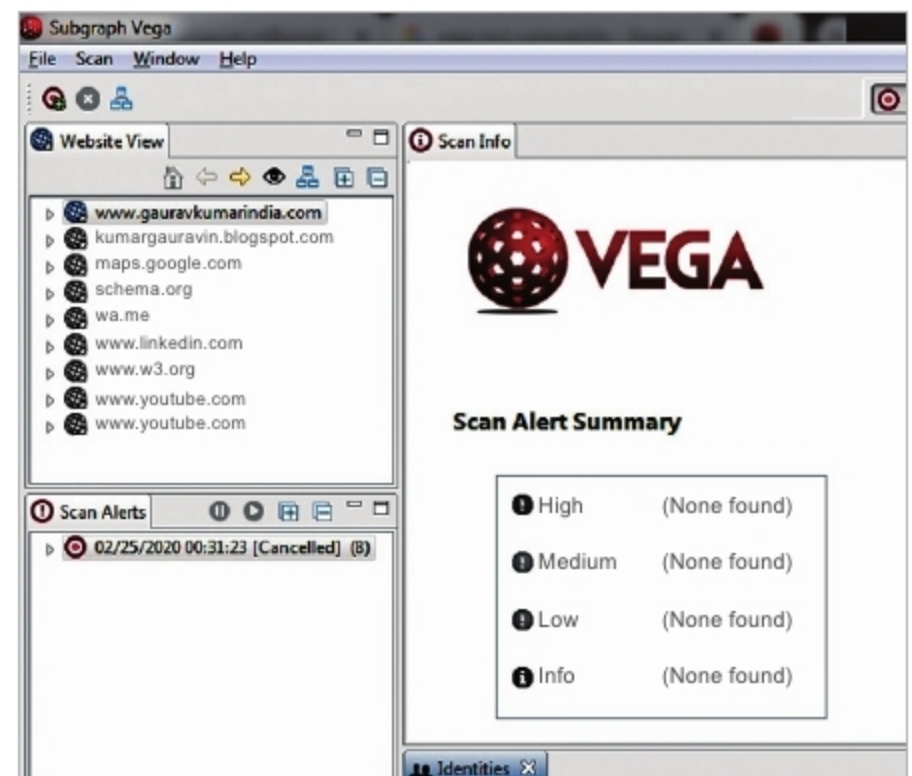


Figure 2: Scan alert summary in Vega

Using the **Android Asset Packaging Tool (AAPT)**, the permissions associated with the APK can be analysed critically so that data breach attempts and the anonymous copying of information can be avoided. In some cases, smartphone users do not download the Android app from the Google Play Store